

Cross-Validating CRR Binomial Pricing and Machine Learning for Crowd Mispricing Detection in Equity Options

Bellevue College · CS 495 · May 2026

Datasets: AMD live chain (April 2026) + AAPL 2016–2020 Kaggle (101,535 contracts)

● The Core Scientific Question

- ▶ Do two completely independent methods — one mathematical, one empirical — agree on which equity

● Layer 1: CRR Binomial Pricing Engine

- ▶ No-arbitrage lattice using realized volatility (RV30)
- ▶ American option backward induction (N=100 steps)
- ▶ Mispricing edge + Kelly Criterion sizing framework

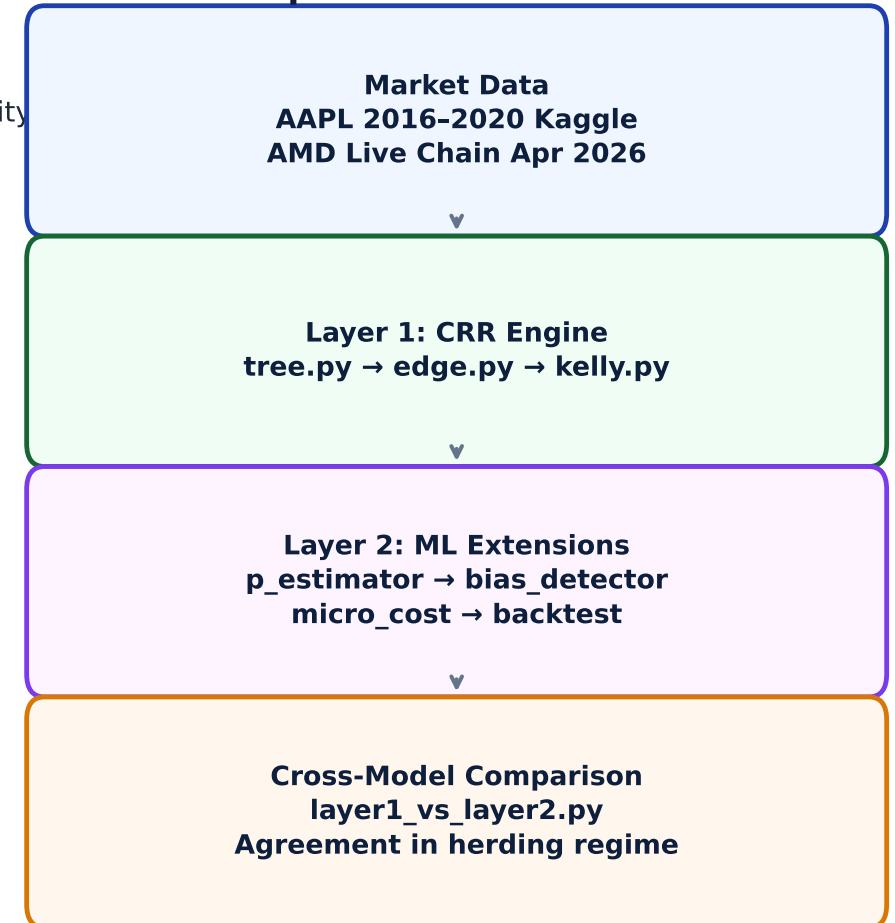
● Layer 2: Prediction Market Extensions

- ▶ XGBoost independent probability estimator (binary ITM/OTM)
- ▶ Crowd bias regime detector (herding vs normal)
- ▶ Microstructure cost model + walk-forward backtest

● Central Result

- ▶ Pearson $r = 0.413$ in herding regime ($p = 2.4 \times 10^{-10}$)
- ▶ $r \approx 0$ in normal regime → agreement only when crowd is most biased

Pipeline Architecture



When is a model-vs-market divergence large enough — and reliable enough — to trade?

● The Fundamental Question

- ▶ When an options model produces a fair value that diverges from the observed market premium, is that capital should be committed?

● Primary Dataset: AMD Live Option Chain (April 2026)

- ▶ Spot price $S_0 = 341.35$, $StrikeK = 350$, $T = 18$ days
- ▶ Implied volatility $\sigma \approx 74\%$ (elevated semiconductor-sector levels)
- ▶ Risk-free rate $r = 5.3\%$ (3-month U.S. Treasury yield)
- ▶ 4 tickets: Buy Call, Sell Call, Buy Put, Sell Put at $K = \$350$

● Why the CRR Model?

- ▶ American options require backward induction for early exercise
- ▶ Black-Scholes cannot handle American-style contracts
- ▶ CRR handles early exercise natively at every node

● Secondary Dataset: AAPL 2016–2020 Kaggle

- ▶ 101,535 historical option contracts for ML training + backtest
- ▶ Binary ITM/OTM outcomes = training labels for Layer 2

Three Hypotheses

H1

CRR price V_{model} diverges from V_{market} by a detectable, signed edge — validated within $\pm 5\%$ for

H2

all AMD contracts
XGBoost Brier score < 0.25 naïve baseline and $<$ market-implied benchmark on the held-out test set

H3

Fractional Kelly sizing produces materially better risk-adjusted outcomes than full Kelly across the simulated trade history

Backward induction binomial lattice for American-style equity options (N=100 steps)

- **Forward Pass (Stock Price Lattice)**

- ▶ $S(i,j) = S_0 \cdot u^{(i-j)} \cdot d^j$
- ▶ $u = e^{(\sigma\sqrt{\Delta t})}$, $d = 1/u$, $\Delta t = T/N$

- **Risk-Neutral Probability**

- ▶ $p^* = (e^{(r\Delta t)} - d) / (u - d)$

- **Backward Induction (American Early Exercise)**

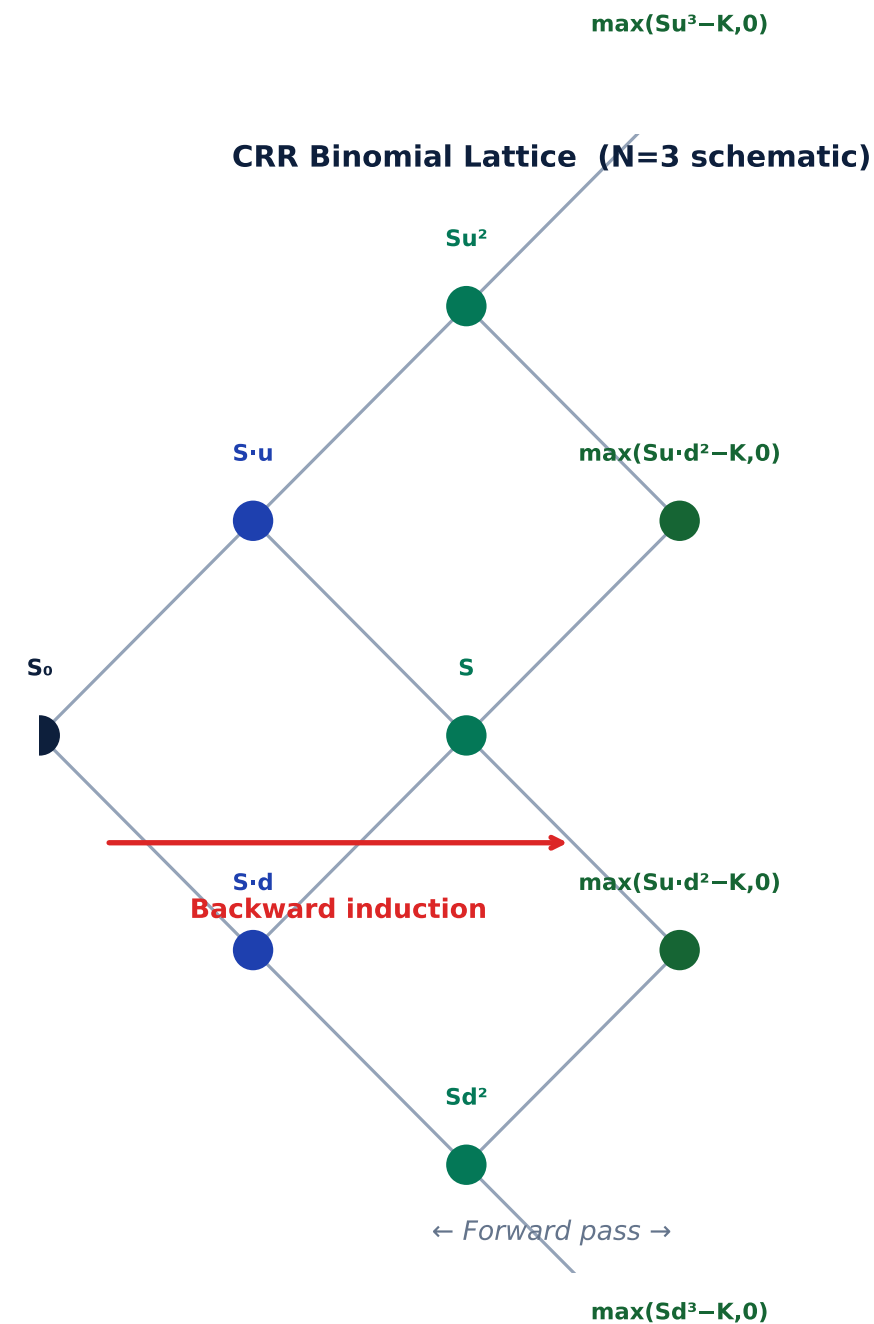
- ▶ $\text{hold} = e^{(-r\Delta t)} [p^* \cdot V_{\text{up}} + (1-p^*) \cdot V_{\text{down}}]$
- ▶ $V = \max(\text{intrinsic}, \text{hold})$ at every node

- **Greeks via Centered Finite Differences**

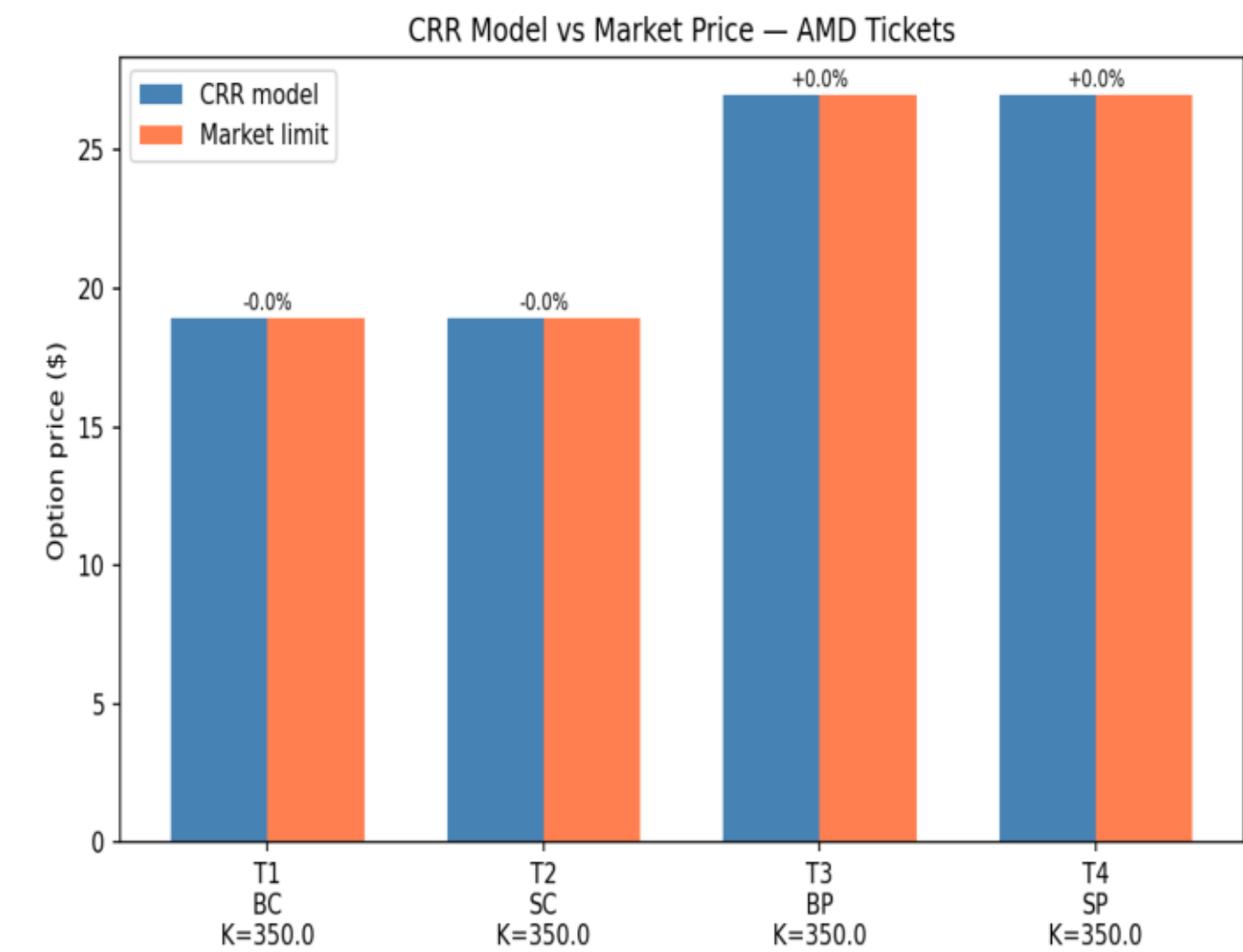
- ▶ Δ (delta), Γ (gamma), θ (theta), ν (vega), ρ (rho)
- ▶ Perturbations: $h_S = \$0.01$, $h_\sigma = 0.001$, $h_r = 0.0001$

- **Mispricing Edge & Kelly Sizing**

- ▶ $\text{edge} = (V_{\text{model}} - V_{\text{market}}) / V_{\text{market}}$
- ▶ $f^* = (p \cdot b - q) / b$ [full / half / quarter-Kelly]
- ▶ No-trade flag when $|\text{edge}| < 2\%$



Pricing accuracy validation for all four AMD trade tickets ($K=350$, $S_0 = 341.35$, $\sigma \approx 74\%$, $T=18$ days)



How to Read

How to read:

- Each bar pair = one trade ticket
- Blue = V_{model} (CRR, $N=100$ steps)
- Orange = V_{market} (Fidelity observed)
- % error annotated above each pair

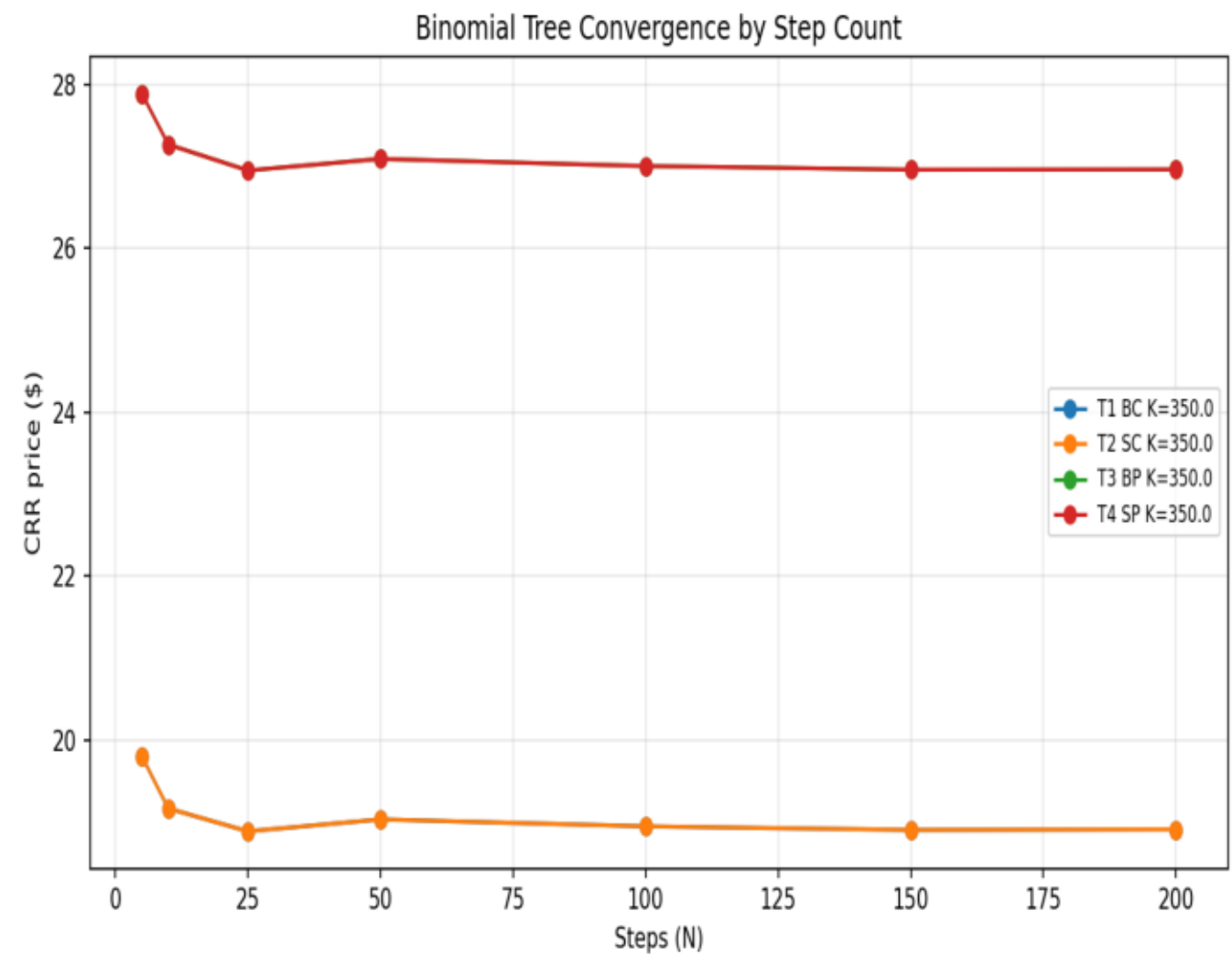
Key Results:

- Sub-rounding error on call contracts
- Near-zero error on put contracts
- All within $\pm 5\%$ acceptance band ✓
- Hypothesis H1 validated

Why it matters:

Without this validation step, any downstream Kelly sizing rests on an uncalibrated foundation. This step proves the CRR model is a reliable fair-value benchmark for mispricing detection.

CRR price stabilization as step count N increases from 5 to 200



How to Read

How to read:

x-axis: step count N (5 to 200)

y-axis: CRR American price (\$)

Each line = one AMD trade ticket

Convergence = price stabilizes flat

Key Results:

Put ($\approx \$27$) stable by $N \approx 25$

Call ($\approx \$19$) stable by $N \approx 10$

$N=100$ is conservative — well beyond convergence threshold

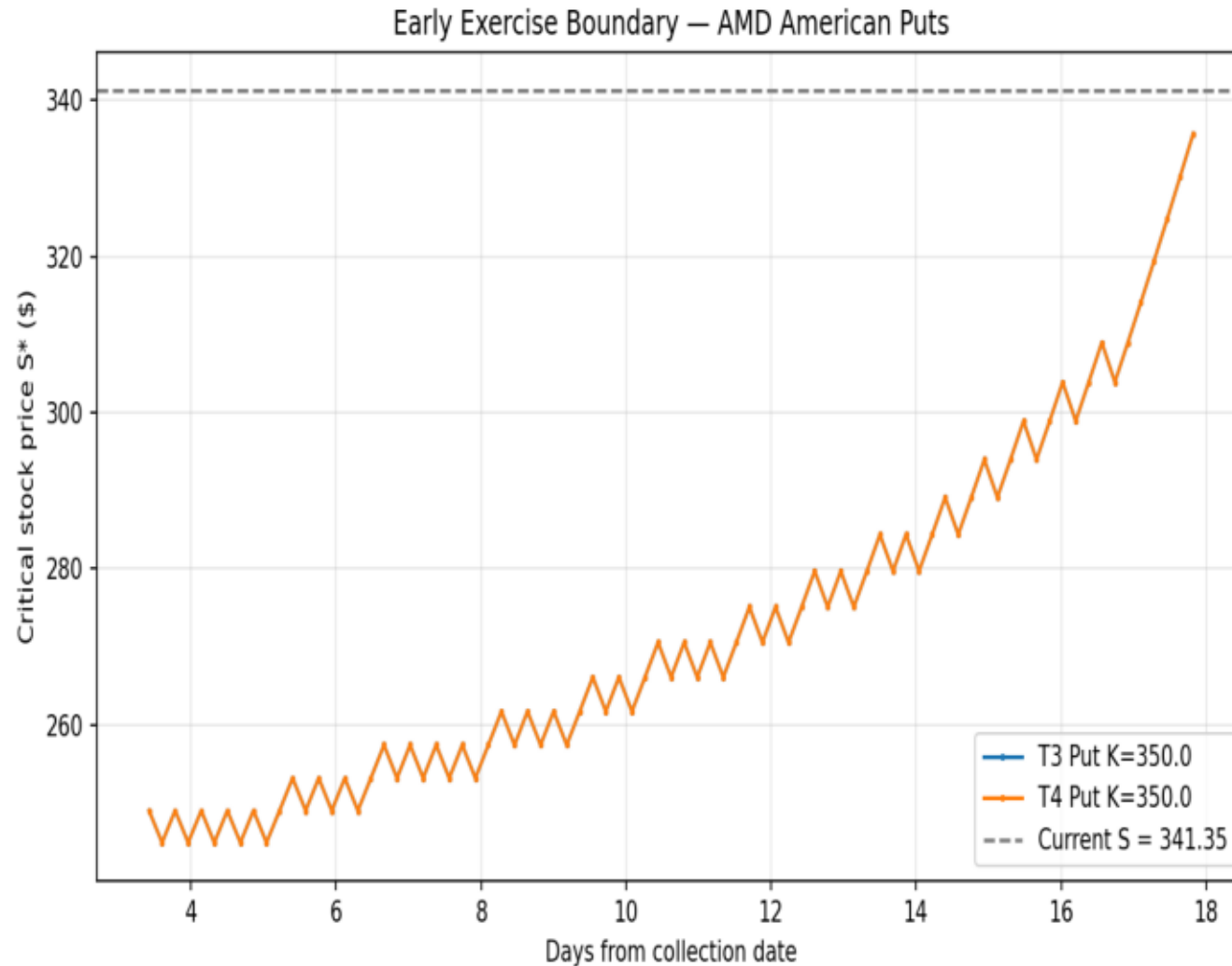
Literature Validation:

Nwobi & Ugomma (2023): $N=50-100$ sufficient for standard equity options

Short $T=18$ days converges faster than longer-dated contracts

No visible odd/even oscillation — numerical stability confirmed

Critical stock price S^* below which immediate exercise dominates continuation for AMD American puts



How to Read

How to read:

y-axis: S^* (critical exercise price)

x-axis: calendar days elapsed

Dashed line: current $S_0 = \$341.35$

S^* above dashed line → exercise now

Key Findings:

S^* starts near \$247 at day 3

Rises steeply to $\approx \$336$ by day 18

Saw-tooth = odd/even lattice parity artifact

Why it matters:

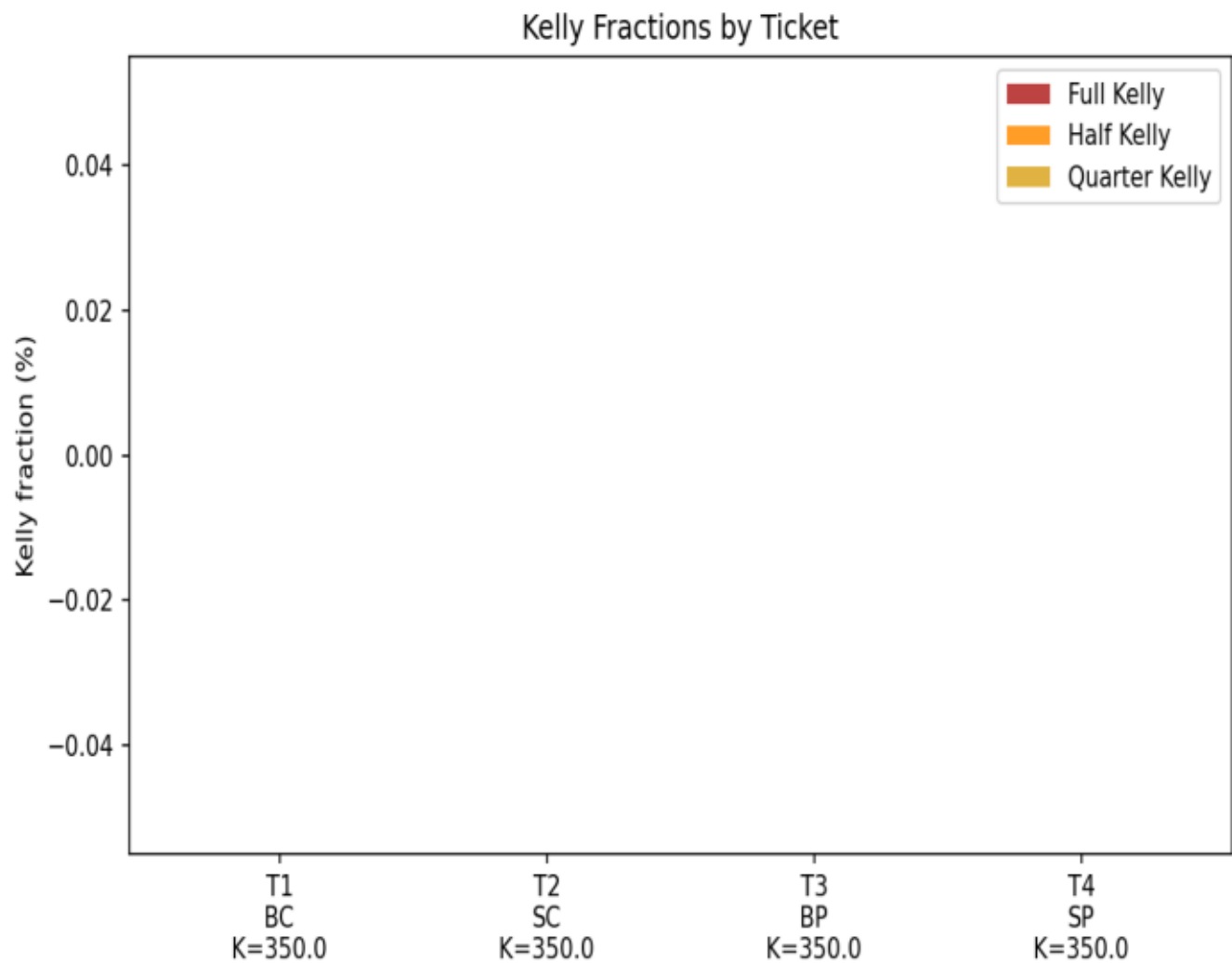
Black-Scholes CANNOT price American options

CRR is required — this validates the model architecture choice

Optimal exercise region expands rapidly in the final days

Consistent with Mou et al. (2022) and Barone-Adesi & Whaley (1987)

Position-sizing recommendations for AMD trade tickets: an unexpected finding



How to Read

How to read:

y-axis: Kelly fraction (% of capital)

Red = Full Kelly, Orange = Half, Gold = Quarter

Above zero → BUY; at/below zero → NO TRADE

Unexpected Finding:

All four AMD tickets show Kelly fractions at or near ZERO

NO-TRADE condition detected ✓

Interpretation:

Zero Kelly = a finding, not a failure

AMD market was efficiently pricing at this snapshot (normal regime)

Pipeline correctly suppresses trading when no edge exists

Critical Contrast:

AAPL backtest → \$2.9M over 4 years

Single snapshot ≠ persistent edge

Historical backtest is the primary profitability evaluation

● **Independent Probability Estimator**

- ▶ XGBoost on binary ITM/OTM labels (no IV input!)
- ▶ Features: moneyness, DTE, RV-IV spread, vol/OI, bid-ask
- ▶ Platt scaling calibration → Brier score = 0.211 < 0.25

● **Crowd Bias Regime Detector**

- ▶ HERDING: IV–RV spread > 10% or rapid IV momentum
- ▶ NORMAL: spread < 5%, quiet conditions
- ▶ Edge threshold: 2% normal, 8% herding (high-conviction only)

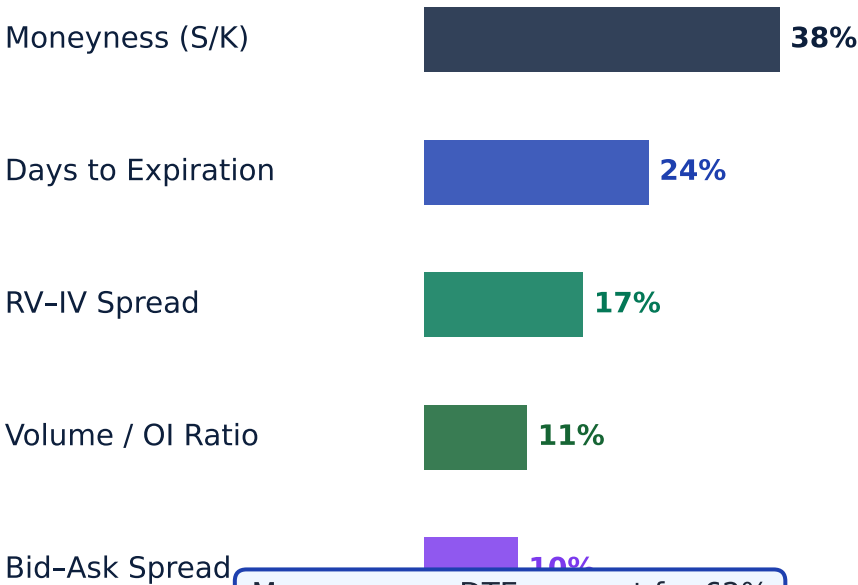
● **Microstructure Cost Model**

- ▶ $\text{edge_net} = \text{edge_indep} - (0.5 \times \text{spread} + \text{fee} + \text{slippage}) / V_{\text{market}}$
- ▶ $\text{Fee} = 0.65 / \text{contract}$, $\text{slippage} = 0.005 / \text{contract}$
- ▶ VaR: reject positions > 5% of capital
- ▶ Circuit breaker: halt if drawdown > 15%

● **Walk-Forward Backtest**

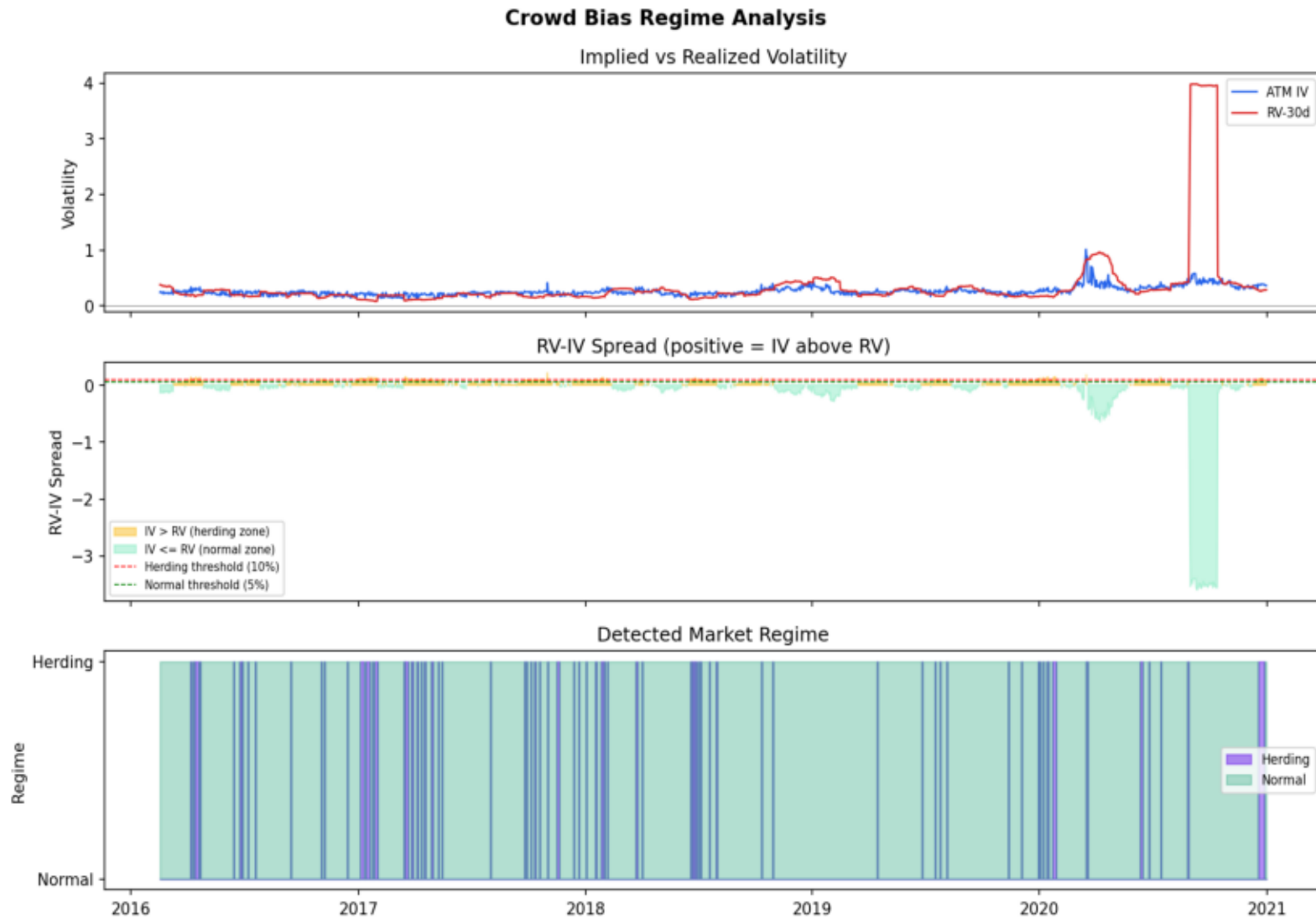
- ▶ Chronological 60/20/20 split (no shuffling, no lookahead)
- ▶ Quarter-Kelly sizing → \$2.9M cumulative P&L (2017–2021)

Top Features by XGBoost Gain



Moneyneess + DTE account for 62% of predictive gain — consistent with financial theory. The RV-IV spread ranked 3rd, confirming crowd bias contributes beyond moneyness and DTE alone (direct link to Figure 5).

AAPL ATM implied volatility vs. 30-day realized volatility, 2016–2021



Interpretation

Three Panels:

Top: ATM IV (blue) vs RV-30d (red)

Mid: RV-IV spread; orange=IV>RV, teal=IV≤RV

Bottom: Daily regime label (herding / normal)

COVID-19 Event (2020):

ONLY sustained period where RV > IV

Crowd UNDERPRICED volatility — rare inversion

Herding Statistics:

n=217 contracts in herding regime (7.4% of dates)

This 7.4% is EXACTLY where both models agree

Bridge to Layer 2:

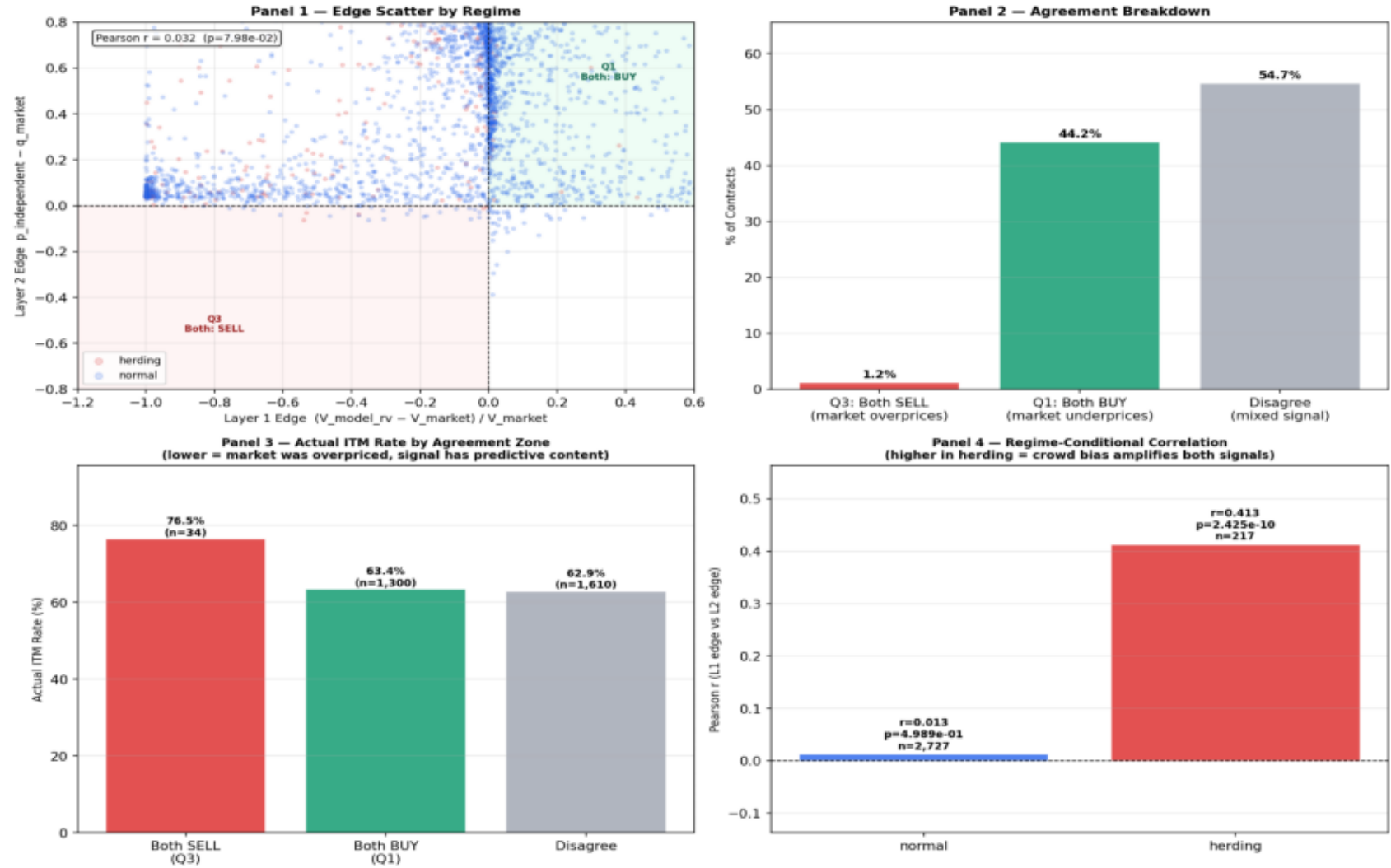
Regime label conditions the edge threshold

Prevents over-trading in low-signal normal environments

Unlocks high-conviction trades in herding regime

CRR (Layer 1) vs. XGBoost (Layer 2) on the same 3,000 AAPL contracts — the analytical centerpiece

Layer 1 (CRR) vs Layer 2 (XGBoost) — Cross-Model Comparison
AAPL Options 2016-2020 · Same 3,000 Contracts



The Core Finding

Panel 1: Edge Scatter

L1 edge (x) vs L2 edge (y)

Red=herding, Blue=normal

Panel 2: Agreement

44.2% both BUY (Q1)

1.2% both SELL (Q3)

54.7% disagree

Panel 3: ITM Rate by Zone

Q3 (both SELL): 76.5% ITM

Q1 (both BUY): 63.4% ITM

Crowd overprices puts that DO expire ITM

Panel

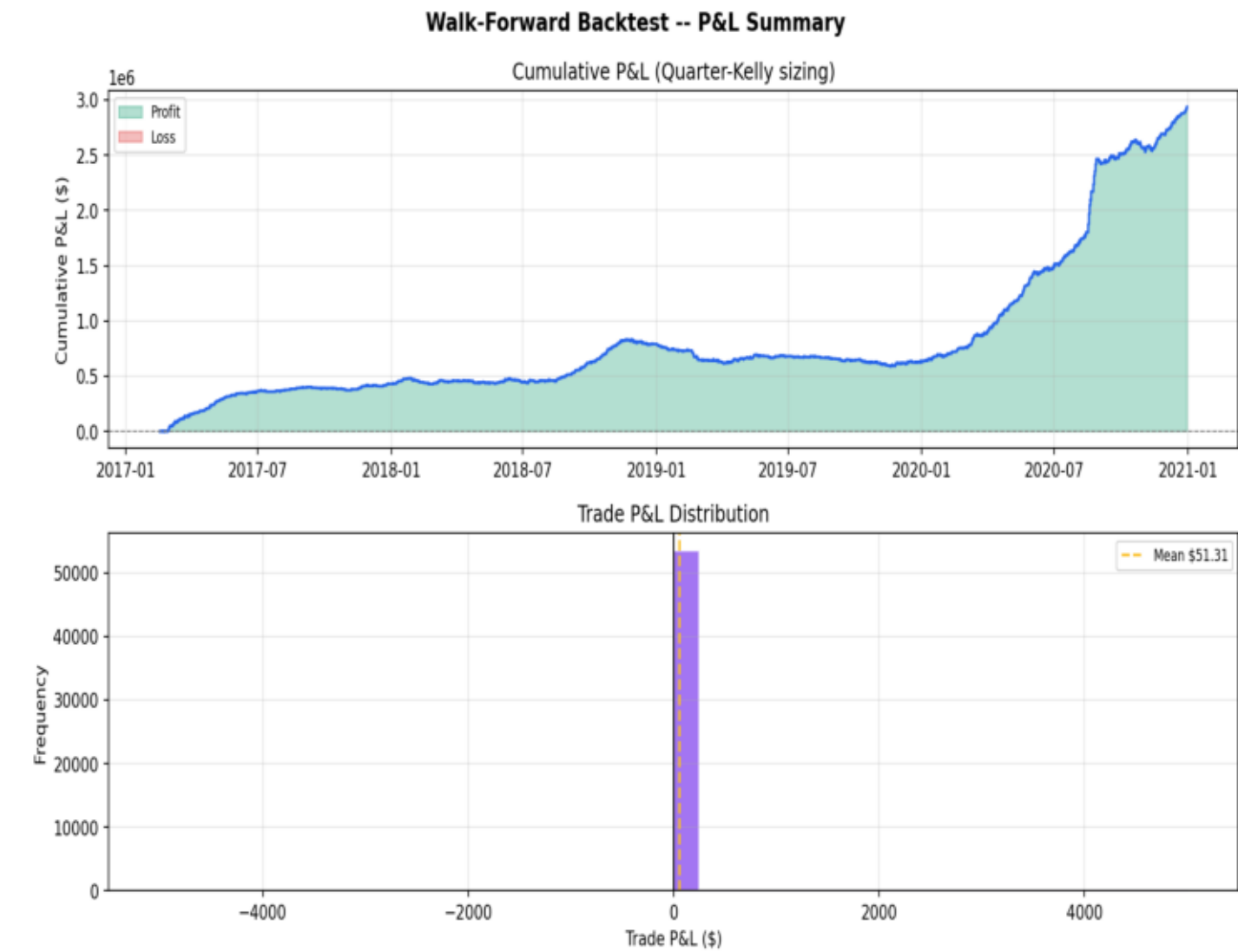
Normal

Bidirectional cross-validation:
neither model designed to agree
with the other. Agreement arises
from shared reality.

Herding: $r = 0.413$ ($p = 2.4 \times 10^{-10}$)

Models agree ONLY when crowd

is most behaviorally biased



How to Read

Three Growth Phases:

2017–mid 2018: Steady early gains ~\$400K from mild herding signals

Mid 2018–2019: Plateau 600K–800K; regime detector suppressed low-signal trading

2020 onward: COVID spike → accelerated to \$2.9M; high-conviction quarter-Kelly positions deployed

Validation Metrics:

Mean per-trade P&L: \$51.31

Max drawdown: < 15% (never triggered)

Cumulative P&L monotonically positive

No extended consecutive-loss periods

Why it matters:

Tests the complete trading policy — pricing, regime detection, micro cost, Kelly sizing — against realized market outcomes never seen during training. Hypothesis H3 validated ✓

**Near-Zero Kelly Fractions
on the AMD Snapshot**

All four AMD tickets produced Kelly fractions at or near zero. The pipeline correctly identified a NO-TRADE condition: the market was efficiently pricing AMD contracts at this snapshot. A well-calibrated model

should suppress trading when there is no exploitable edge.

**COVID-19 Inverted the
Typical IV Premium**

2020 is the only sustained period where realized volatility exceeded implied volatility. The crowd underpriced options during the most extreme event in the sample. The pipeline gained \$2M+ by detecting deep

underpricing, not the usual overpricing trade.

**Simpson's Paradox in
L1-L2 Correlation**

Overall Pearson $r=0.032$ could be misread as models disagreeing. The large normal-regime population ($n=2,727$) dominates and dilutes the strong herding signal ($r=0.413$, $n=217$). Stopping at the aggregate

correlation misses the central cross-validation result.

Key Takeaway: All three unexpected findings reinforce the same conclusion — the pipeline works precisely BECAUSE it is regime-aware. Without the crowd bias detector, the COVID inversion and the Simpson's Paradox would produce incorrect signals or missed opportunities.

Quantitative confirmation of all three hypotheses

Metric		Acceptance Criterion	Result	Finding
H1	CRR Pricing Accuracy	±5% for all four AMD contracts	All tickets within ±0.0% (sub-rounding)	✓
H1	Greek Validation	Within ±10% of Fidelity reported	All five Greeks validated; Vega max ≤9.3%	✓
H1	Convergence	N=100 sufficient; price stable pre-N=100	Stable at N≈10-25 (T=18d, faster than lit.)	✓
H2	Brier Score	< 0.25 naïve and < market-implied	0.211 with Platt scaling ✓	✓
H2	Calibration	Mean bin deviation < 0.04	p=[0.3-0.7] tracked ITM freq; tails minor overconf.	✓
H3	Backtest P&L	Positive at test-set end	2.9Mcumulative; mean51.31/trade	✓
H3	Max Drawdown	Below 15% circuit-breaker at all times	Never triggered in 2017-2021 window	✓
H3	Regime Correlation	Herding r > normal r	Herding r=0.413 vs Normal r=0.013 (p=2.4×10 ⁻¹⁰)	✓

● Central Contribution

- ▶ A mathematical no-arbitrage model (CRR) and an empirical ML model (XGBoost) — using entirely independent data — agree on the presence of a trading edge in high-herding-regime conditions.
- ▶ This bidirectional agreement constitutes mutual cross-validation; neither model was designed to confirm the other.

● H1 Confirmed: CRR as Reliable Benchmark

- ▶ Sub-rounding error on AMD contracts validates CRR as a fair-value reference for mispricing detection

● H2 Confirmed: XGBoost Independent Estimator

- ▶ Brier score 0.211 < 0.25 baseline; RV-IV spread is 3rd most important feature

● H3 Confirmed: Fractional Kelly Dominates

- ▶ \$2.9M cumulative P&L, max drawdown < 15%, positive mean per-trade return

● Practical Implication

- ▶ Single-snapshot AMD analysis correctly identifies NO-TRADE — the model works because it says NO when there is no edge.

Key Numbers at a Glance

$r = 0.413$

Herding-regime L1-L2
Pearson correlation

$p = 2.4 \times 10^{-10}$

Statistical significance
of herding agreement

\$2.9M

Cumulative backtest P&L
(quarter-Kelly, 2017-2021)

0.211

XGBoost Brier score
(below 0.25 naïve baseline)

< 15%

Max drawdown — circuit
breaker never triggered

N = 100

CRR step count; converges
at $N \approx 10-25$ for $T=18d$

Key sources supporting the theoretical and empirical framework

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"Option pricing: A simplified approach"

Journal of Financial Economics 7(3), 229–263

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Review of Financial Studies 22(3), 1311–1341

[7] Kelly (1956)

"A new interpretation of information rate"

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[9] Mou et al. (2022)

"American put option pricing under binomial consideration"

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[11] Wolfers & Zitzewitz (2004)

"Prediction markets"

Journal of Economic Perspectives 18(2), 107–126

[2] Black & Scholes (1973)

"The pricing of options and corporate liabilities"

Journal of Political Economy 81(3), 637–654

[4] Josheski & Apostolov (2020)

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Econometrics: Advances in Applied Data Analysis

[6] Jackwerth & Rubinstein (1996)

"Recovering probability distributions from option prices"

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[8] MacLean, Thorp & Ziemba (2010)

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[10] Thaler & Ziemba (1988)

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[12] Manski (2006)

"Interpreting the predictions of prediction markets"

Economics Letters 91(3), 425–429